



June 17, 2026

Dear Admissions Committee,

I am writing to express my strong interest in pursuing a PhD as a Research Associate / PhD Student in Information Theory of Sensing and Coordination at the Technical University of Munich (TUM) in Munich, Germany. My name is Ismail Aissaoui, and I am an AI Data Science Engineer with a passion for developing resilient communication systems.

As a holder of a master's degree in Engineering, I have acquired a solid foundation in Information Theory, which aligns well with the requirements of this PhD program. Throughout my academic and professional journey, I have been fascinated by the potential of AI to revolutionize various fields, including sensing and coordination. My research experience has equipped me with strong skills in Information Theory, and I am eager to leverage these skills to contribute to the development of resilient communication systems.

One achievement that resonates particularly well with this PhD program is my work on *MedicalMambaTS*, a linear – time state space model for sub – millisecond bedside inference. This project highlights my ability to develop fast and efficient algorithms that leverage the time processing capabilities of *MedicalMambaTS* enabled devices to generate a common approximation.

I am confident that my academic background, research experience, and achievements make me an excellent candidate for this PhD program. I am excited about the opportunity to join the TUM community and contribute to the development of innovative solutions in Information Theory of Sensing and Coordination.

Sincerely,